We estimate θ by matching the predicted marginal $F_{\boldsymbol{\theta}}$ to the observed F_{obs} using a distributional distance, here the Wasserstein-1 (earth mover) metric on $v \in [0, 1]$.

For a Beta regime parameterization $P \sim \text{Beta}(\kappa m, \kappa(1 - m))$ (mean m and concentration κ), we evaluate the distance on a coarse grid over (m, κ) and refine locally (e.g., L-BFGS-B).

The heatmap shows $\log_{10}$ distance: lighter regions indicate regime parameters that reproduce the observed current-grade distribution more closely. The marked point is the minimizer $\hat{\theta}$.